MINJE PARK

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Scholar	Google Scholar (Minje Park)

Education

Sungkyunkwan University *Feb 2021 – Feb 2027 (Expected)*

B.S. in Physics, Double Major in Electronic & Electrical Engineering

Research Experience

Generative AI for Cosmological Simulation and Super-Resolution Jun 2026 – Present

Supervisor: Ji-Hoon Kim, Department of Physics and Astronomy, Seoul National University

- Developing generative AI models for cosmological simulation and super-resolution.

Generative Modeling for Multifield Cosmological Simulation May 2025 – Present

Independent Research

- Developed GENESIS, a flow-matching emulator for multifield cosmological generation using CAMELS simulation data.
- Investigating training-free astrophysical and cosmological parameter inference via maximum-likelihood estimation using GENESIS.

Transformer-Based Reconstruction Model for SWGO Jun 2024 – Feb 2026

Supervisor: Ian Watson, Department of Physics, University of Seoul

- Investigated position embedding strategies for transformer-based primary-particle reconstruction model in SWGO.
- Developed flow matching models to synthesize SWGO detector signals from simulation data.

Generative AI for IceCube and SWGO May 2024 – Feb 2026

Supervisor: Chang Dong Rho, Department of Physics, Sungkyunkwan University

- Developed diffusion models and Mixture Density Networks for synthesizing PMT signals for SWGO Tank Simulation.
- Built flow & diffusion models to generate neutrino event in IceCube Observatory.

CNN and Vision Transformer for IceCube Camera Analysis Aug 2023 – Sep 2025

Supervisor: Carsten Rott (University of Utah & Sungkyunkwan University)

- Developed CNN and Vision Transformer models for the IceCube Upgrade Camera System; results published as a poster and proceedings paper at ICRC 2025 (arXiv:2507.18525).

Publications

Performance Study of the IceCube Upgrade Camera System

*C. Rott, **M. Park**, M. Jansson, G. Iverson, S. Choi (for the IceCube Collaboration)*

Proceedings of Science (PoS), ICRC2025; arXiv:2507.18525

Conference Presentations

Apr 2026 — [Talk] **Generative AI for IceCube Observatory**
103rd Spring Meeting of the Korean Physical Society, Daejeon, Republic of Korea

Apr 2026 — [Poster] **Mixture Density Networks for Fast Detector Response Simulation in SWGO**
103rd Spring Meeting of the Korean Physical Society, Daejeon, Republic of Korea

Feb 2026 — [Poster] **Investigation of IGZO-Based 3D Vertical Channel Transistors and Physics-Informed Neural Network Modeling for Next-Generation DRAM**
The 33rd Korean Conference on Semiconductors, Jeongseon, Republic of Korea

Oct 2025 — [Poster] **Diffusion and Flow Matching Model for IceCube Neutrino Event Simulation and Reconstruction**
102nd Fall Meeting of the Korean Physical Society, Gwangju, Republic of Korea

Oct 2025 — [Poster] **Generative AI for Cosmological Simulation and Manifold Representation**
KPS Undergraduate Physics Research Competition, 102nd Fall Meeting of the Korean Physical Society, Gwangju, Republic of Korea

Jun 2025 — [Poster] **Diffusion and Flow Matching Model for IceCube Neutrino Event Simulation and Reconstruction**
4th K-Neutrino Symposium, Seoul, Republic of Korea

Apr 2025 — [Poster] **Modeling the Relationship between PMT Response and Air-Shower Particles for SWGO**
101st Spring Meeting of the Korean Physical Society, Daejeon, Republic of Korea

Oct 2024 — [Poster] **Development of a Deep Learning Model for Analyzing Antarctic Ice Optical Properties**
KPS Undergraduate Physics Research Competition, 101st Fall Meeting of the Korean Physical Society, Yeosu, Republic of Korea

Sep 2024 — [Talk] **Deep Learning for Ice-Property Measurement Using Simulation Images**
IceCube Collaboration Meeting, Madison, Wisconsin, USA

Jul 2024 — [Talk] **Ice-Property Measurement Using a Camera System in the IceCube Upgrade**
3rd K-Neutrino Symposium, Gwangju, Republic of Korea

Apr 2024 — [Poster] **Study of Optical Properties in Antarctic Ice Using Simulated Images**
100th Spring Meeting of the Korean Physical Society, Daejeon, Republic of Korea

Honors & Awards

Best Presentation Award
The 33rd Korean Conference on Semiconductors, 2026
Awarded for poster presentation: "Investigation of IGZO-Based 3D Vertical Channel Transistors and Physics-Informed Neural Network Modeling for Next-Generation DRAM"

Best Presentation Award
4th K-Neutrino Symposium, 2025
Awarded for poster presentation: "Diffusion and Flow Matching Model for IceCube Neutrino Event Simulation and Reconstruction"

Merit Award
Korean Physical Society Undergraduate Physics Research Competition, 2024
Recognized for research on CNN and Vision Transformer models for the IceCube Upgrade Camera System

Skills

Programming	Python, NumPy, SciPy, Matplotlib
Tools	Linux, HTCondor, Git, Bash/Shell, HDF5
Frameworks	PyTorch, MATLAB, JAX
Architectures	Transformer, Diffusion Transformer (DiT), Linformer, Vision Transformer (ViT), Swin Transformer, CNN, ResNet, U-Net, Perceiver
Models	VAE, Diffusion Models, Flow Models, Mixture Density Networks, Physics-Informed Neural Networks, Masked Autoencoder (MAE)
Methods	Optimal Transport Flow Matching, Rotary Positional Embedding (RoPE), Transformer Scaling Laws

Collaboration Memberships

IceCube Collaboration *Aug 2023 – Present*
A neutrino detector embedded in Antarctic ice at the South Pole

SWGO Collaboration *Jun 2024 – Present*
A next-generation ground-based gamma-ray observatory planned for the high-altitude Andes